



OPEN Psychometric properties of the SMART tool for infant oral feeding skill assessment

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Premature and medically complex infants are at risk of feeding problems, specifically the delay in the development of independent oral feedings, which leads to prolonged hospital stays, higher costs, malnutrition, developmental delays, and family stress. Available oral feeding skill assessment tools lack strong psychometrics, feasibility, and cost-effectiveness. We developed a novel tool, the SMART tool, to address these barriers. The objective of this study was to establish the psychometric properties of the SMART tool for assessing infant oral feeding skills. The study design was a prospective, observational, multicenter trial. To establish criterion validity, we hypothesized that the performance of the SMART tool would be similar to that of a comparator tool, the Neonatal Eating Outcome Assessment (NEOA) tool. A convenience sample of 76 infants across four level-3 Neonatal Intensive Care Units were enrolled, and 136 complete feeding observations were included. Compared to NEOA, the SMART tool had a sensitivity of 0.76 (95% CI 0.63–0.89) and a specificity of 0.82 (95% CI 0.72–0.93), with a strong correlation ($r = 0.7$, $p < 0.001$). Inter-rater reliability of post-feed scores was 100% agreement. The SMART tool is valid, reliable, and feasible for assessing infant oral feeding skills.

Keywords Infant nutrition, Feeding skills, Psychometric, Breastfeeding

Abbreviations

AUC	Area under the curve
COSMIN	Criteria: Consensus-based Standards for the Selection of Health Status Measurement Instruments Criteria
CVI	Content Validity Index
CVR	Content Validity Ratio
NEOA	Tool: Neonatal Eating Outcome Assessment Tool
SMART	Tool: State of arousal, Motor tone, Autonomic instability, Response to stimulation, and Total oral skills Tool
SME	Subject Matter Expert

The American Academy of Pediatrics policy on infant hospital discharge, initiated in 1998, includes two nutritional assessment criteria: a sustained pattern of weight gain of sufficient duration and competent oral feeding skills without cardiorespiratory compromise^{1,2}. While weight gain assessment is quantitative, easily measurable, and depends on caloric and volume intake, feeding skill assessment (FSA) usually remains a qualitative discussion between clinicians. The inherent qualitative nature of FSA makes it challenging to measure, and no gold-standard assessment tool is currently available for clinical use. This introduces immense variation in communication among clinicians (physicians, nurses, speech-language pathologists, lactation consultants, nurse practitioners, and other staff) and families, which may lead to changing care plans and inconsistent feeding practices. For preterm and medically complex infants in the Neonatal Intensive Care Unit (NICU), this is particularly problematic. Acute and chronic feeding issues are well-documented in the literature and include prolonged length of stay, hospital readmission, family stress, and risk of ongoing feeding difficulties post-discharge^{3–7}. In 2019, our health system (Advocate Health) identified the need for a standard FSA tool for our 16 NICUs in the Midwest region.

Literature review

The literature review was guided by five criteria in the search for an optimal FSA tool, which included: (1) application: use for infants (term and premature) with different manners of oral intake (breast and bottle), (2) holistic: comprehensive in assessing the many facets of feeding, (3) interpretability: provides a score for

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interpretation regarding safety and quality of feeds that can be communicated between clinicians and families, (4) robust psychometric properties for validity and reliability and (5) feasible: practical, readily accessible and efficient for bedside clinicians to complete training on the tool, perform scoring, interpret results, document in the electronic health record and at a low cost to the health system. None of the available tools in our literature review met all of these criteria (Supplementary Figure S1)^{8–12}. *Currently, no gold-standard FSA tool is available for bedside clinicians*. According to research, available FSA tools should be used with caution^{8,13}. All NICU clinicians need a valid and reliable assessment tool that is highly feasible to objectively evaluate infant feeding skills beyond the volume of oral intake.

Need for FSA tool

Psychometric properties refer to a clinical tool's validity, reliability, and feasibility. Clinical tools with poor reliability, validity, or feasibility may be unfavorable for patients¹⁴. Measurement of the psychometric properties of any tool is essential before implementation into clinical care and evaluation of clinical outcomes. In the NICU context, the goal was to evaluate different FSA tools with numeric scores that elucidate the quality and safety of infant feedings. Three existing tools with scores were identified as possible options. The three FSA tools were the Infant Driven Feeding Scale (IDFS)¹⁵, the Early Feeding Skills Assessment Tool (EFS)^{9,16}, and the Neonatal Eating Outcome Assessment (NEOA)^{10,17}. IDFS was not considered further due to limited content validity using the consensus of experts (the Delphi method)¹⁸. Newer research has shown IDFS and EFS have cross-cultural validation with translations into non-English languages. The IDFS tool has been translated and validated in the Turkish language¹⁹. The EFS tool has been translated and validated into many languages, including Spanish, Turkish, Portuguese, Persian, and Indonesian^{20–24}. EFS and NEOA have strong psychometric properties with established validity and reliability, and are organized to identify the deficit areas in FSA more easily to facilitate domain-specific intervention^{10,16,17}. However, both tools have poor feasibility because they are complex for the bedside clinician, time-consuming to administer, and not integrated into electronic health records. The EFS tool is copyrighted. The EFS Tool research version has been psychometrically tested; however, it lacks a pre-feed assessment section and a score-based clinical decision tree. The NEOA has been used for detailed feeding assessments by therapists. However, the complexity of the tool makes it challenging for a bedside clinician to use it in daily practice. The lack of integration of both tools into electronic medical records to calculate and document scores and develop actionable tables is another significant hindrance to easy adoption. Without an adequate FSA tool that could satisfy the selection criteria, a novel FSA tool called the SMART tool was designed to meet the needs of NICU clinicians, providing an objective measure of feeding readiness, quality, and safety. The objective of this study was to establish the psychometric properties (validity, reliability, and feasibility) of the SMART tool for infant oral feeding skill assessment. To establish criterion validity, we hypothesized that the performance of the SMART tool would be similar to that of a comparator tool, the Neonatal Eating Outcome Assessment (NEOA) tool.

Methods

SMART tool design

Ravi Mishra, MD, and Anne Albi, SLP/IBCLC, are the co-creators of the SMART tool, a novel infant FSA tool. The SMART tool evaluates feeding skills across five domains: state of arousal, motor tone, autonomic instability, response to stimulation, and total oral skills. These domains form the acronym of the SMART tool. All five domains are conducted before (pre-feed) and after each feeding (post-feed). The Total Oral Skills (TOS) is assessed with different measures in the pre- and post-feed sections. In the pre-feed setting, TOS refers to the duration of non-nutritive suck. In the post-feed setting, TOS refers to nutritive suck performance as measured by five sub-items, which include rooting, sucking bursts, bolus control, suck swallow breathing coordination, and stress cues. The three tools, EFS, NEOA, and SMART, assess similar components of the Total Oral Skills; however, each tool differs in scoring methods. The SMART Tool considers Total Oral Skills as one of the five factors. In contrast, the EFS and NEOA consider each component of Total Oral Skills as an independent factor.

The Synactive Theory of Development, developed by Heidelise Als, provides the theoretical foundation for the design of the SMART tool²⁵. Synactive means “together in action” and refers to infant development using interactions. There are continuous interactions within subsystems (State, Motor, Autonomic, and Attention), continuous interaction of subsystems with the whole organism (infant), and continuous interactions between the infant and its environment, including the parents and care providers. These multilayered, continuous interactions lead to infant development. Understanding these interactions provides a window of opportunity to influence infant development. This view of development has been defined as *Synactive*, since at each stage in development and at each moment of functioning, the various subsystems discussed are existing side by side, often truly interactive or in a relative supporting, holding pattern, as if providing a steady substratum for one of the system's current differentiation (synaction)²⁶. It also aligns with trauma-informed care principles, the neonatal integrative developmental care model, life course health development intervention, and infant-family-centered developmental care^{27–30}.

The SMART tool (Table 1) is practical for bedside clinicians due to its simplified scoring, fast and intuitive features, and clinical support decision tree for safe feeding. It includes pre-feed and post-feed sections, with a total score of 100 possible points for each, where higher is better for standardizing interpretations. For feeding safety, clear instructions were included as “Hard Stops”, which help identify signs of severe autonomic instability and prevent aspiration or choking (Table 1, Section C). This is a unique feature of the SMART tool, and impacts infant safety and end-users' decision-making process. A decision tree (Table 1, Section E) is included in the SMART tool, which helps guide the initiation of oral feeding and qualify feeding skill levels. Oral feeding should only be offered if a pre-feed score is 70 or more. The post-feed score is to classify the feeding skill level into three

A. PRE-FEED evaluation	Severe	Moderate	Mild	None	Non-Nutritive
Assess Impairment	5	10	15	20	Write Score
State of Arousal	Sleepy	Variable	Arousable	Alert	
Motor tone	Hyper/Hypotonic	Fluctuating	Reduced flexion	Normal flexion	
Autonomic Instability [See C]	Severe	Moderate	Mild	None	
Response to Stimulation	None	Inconsistent	Disorganized	Normal	
Total Oral Skills (seconds)	0 to 15	15 to 30	30 to 60	Above 60	
Pre-Feed Score Total					
B. POST-FEED evaluation	Severe	Moderate	Mild	None	Nutritive
Assess Impairment	5	10	15	20	Write Score
State of Arousal	Sleepy	Variable	Arousable	Alert	
Motor tone	Hyper/Hypotonic	Fluctuating	Reduced flexion	Normal flexion	
Autonomic Instability [See C]	Severe (Hard stop)	Moderate	Mild	None	
Response to Stimulation	None	Inconsistent	Disorganized	Normal	
Total Oral Skills [See D]	Write the sub-total of Total Oral Skills from Section D here.				
Post-Feed Score Total					
C. Autonomic instability	Severe-Hard stop	Moderate	Mild	None	Note lowest score
Lowest Heart Rate (rate/min)	60	80	100	120	Vitals on Monitor
Lowest Oxygen Saturation (%)	60	70	80	90	
Highest Respiration (rate/min)	100	90	80	70	
Apnea (20 s or more)	1	0	0	0	Visceral Events
Cough or choking events	2	1	0	0	
Number of Events	3	2	1	0	
D. Total oral skills (Nutritive)	Severe	Moderate	Mild	None	
Assess Oral Impairment	1	2	3	4	Write Score
Rooting	No response	Diminished	Sometimes	Always	
Sucking bursts	0 to 3 sucks	3 to 5 sucks	5 to 10 sucks	10 to 30 sucks	
Bolus Control	Cough/Choke	Disorganized	Spillage	Normal	
Suck Swallow Breathing	Hard Stop	Needs pacing	Uncoordinated	Coordinated	
Stress Cues	Shut Down	Frequent	Sometimes	None	
Sub-total of Total Oral Skills					
E. Assess skill level	Pre-feed score	Decision	Action	Post feed score	Skill level
	Under 70	No oral feed	Only Tube Feed	Under 60	Caution
Decision Tree			Non-Nutritive suck	60 to 90	Developing
(Scores have no units)	70 or more	Oral feed		Above 90	Capable
F. Non-scoring items	Mother care	Infant Oral-motor	Position - Baby	Breastfed nipple	Interventions
Birth Gestational Age	Involved	Cleft lip/palate	Cradle	Breastfed	Burping
Post Menstrual Age	Breastfed/Pump	Recessed Jaw	Cross Cradle	Use of Shield	Change Nipple
Growth Failure	Skin to Skin Care	Tongue large	Face-to-Face	Breastmilk stored	Change Position
	Breast issues	Tongue tie	Football	Tube Fed	Non-Nutritive
Infant fed by	Milk Supply Low	Syndromes	Side-lying (baby)		Pacing
Baseline	Mother issues	Infant issues	Position - Mom	Bottle-fed nipple	Interventions
Date	Diabetes	Cardiac	Laid-back (recline)	Ultra-Preemie	Re-alerting
Time	Hypertension	Respiratory	Side-lying (mom)	Preemie	Reduce Stimuli
Heart Rate	Mental	Neurological		Slow Flow	Rest Breaks
Respiratory Rate	Physical	Infection		Regular	Swaddle
Saturations	Social	Surgical		Other	Others
Oxygen support	Medication	Medication			

Table 1. SMART tool.

categories: Caution (under 60), Developing (60 to 90), and Capable (above 90). The SMART tool comprises a glossary of terms (Table 2) and defined procedures to guide care.

The SMART tool was shared with a system-wide workgroup of physicians, nurses, speech-language pathologists, and lactation consultants to solicit feedback. The authors established the score cut-offs for the clinical support decision tree. The subject matter experts (SMEs) (Lactation Consultants and Speech Language Pathologists) were selected from all four sites, having more than five years of experience in infant feeding at the bedside, who assisted in testing the tool. The SMEs participated in the content validity testing.

1. NON-NUTRITIVE SUCKING (NNS): any suckling when fluid is not being introduced via the mouth.
2. NUTRITIVE SUCKING (NS): any suckling where the fluid is being introduced, and bolus formation is occurring in the mouth and throat.
3. STATE OF AROUSAL: Infant's condition during stimulation by the caregiver
3.1. Sleepy: Light or deep sleep state of arousal, despite stimulation
3.2. Variable: Alternating between sleepy and alert states
3.3. Arousable: Transition to a more alert state with handling by the caregiver
3.4. Alert: Awake and attentive to environmental stimulation
4. MOTOR TONE: Posture, tone, and movement in the body to aid postural alignment.
4.1. Hypotonic: relaxed body tone, limp, giving up.
4.2. Hypertonic: increased body tone, rigid, arching, hyperextensions of limbs, locked extremities
4.3. Fluctuating: Alternating between a relaxed tone and an increased tone, diffuse
4.4. Reduced Flexion: bending of the body towards the midline, as in hands to midline, though reduced.
4.5. Normal Flexion: Active Flexion with alertness
5. AUTONOMIC INSTABILITY: Changes in unconscious bodily functions, including respiration, heart rate, oxygenation, and visceral events.
6. RESPONSE TO STIMULATION: Demonstrate feeding readiness cues with stimulation. It assesses one of the domains of attention-interaction behavior under the Synactive theory. Here, one aspect of feeding, "rooting," and its impact on attention-interaction is given an individual domain level. Rooting is also assessed as a sub-domain item under Total Oral Skills.
6.1. None: Absent rooting and jaw opening in response to stimulation. Baby remains uninterested in feeds and may shut down.
6.2. Inconsistent: Rooting and/or jaw opening, though not on each presentation. It may require multiple repetitions of stimulation to elicit a response.
6.3. Disorganized: Rooting and/or jaw opening in response to stimulation, but disorganization noted (i.e., hyper-rooting, failure to latch promptly)
6.4. Normal: Immediate, consistent rooting reflex elicited with stimulation near the mouth. This is associated with a smooth, well-modulated posture and a well-regulated tone.
7. TOTAL ORAL SKILLS (NNS): SUCKING BURSTS: Series of consecutive sucks before a pause. If bottle feeding, offer the infant a gloved finger or pacifier, wait for rooting and acceptance, and then calculate the duration the infant can sustain NNS before disengagement. If breastfeeding, assess the first 60 s of suckling at the breast to evaluate NNS.
8. TOTAL ORAL SKILLS (NS): ROOTING RESPONSE: The reflexive turning of the head and opening of the mouth in response to stimulation near the lips.
8.1. No response (to stimulation): Absent rooting and jaw opening in response to stimulation.
8.2. Diminished (rooting): Slow elicitation of the rooting reflex following multiple repetitions of stimulation.
8.3. Sometimes (rooting): Inconsistent rooting and/or jaw opening in response to stimulation.
8.4. Always (rooting): Consistent, immediate rooting reflex elicited with stimulation near the mouth.
9. TOTAL ORAL SKILLS (NS): SUCKING BURSTS: Series of consecutive sucks before a pause, with pacing as needed.
10. TOTAL ORAL SKILLS (NS): BOLUS CONTROL: Containment of food or liquid in the mouth during the oral phase and maintained through the pharyngeal phase of the swallow.
10.1. Cough/Choke: Coughing/Choking with disruption of breathing associated with feeding.
10.2. Disorganized: Decreased management of bolus in the pharyngeal phase of swallow characterized by audible swallows, occasional stridor, and/or multiple swallows
10.3. Spillage: Anterior loss of bolus from lips
10.4. Normal (bolus control): Quiet swallows with no loss of fluid at the lips
11. TOTAL ORAL SKILLS (NS): SUCK, SWALLOW, BREATHE (SSB) COORDINATION: Rhythmic sequence of sucking/swallowing/breathing.
11.1. Hard Stop:
11.1.1. Vitals changes: heart rate ≤ 60 /minute, oxygen saturations $\leq 60\%$, respiration rate ≥ 100 /minute
11.1.2. Visceral events: one apnea event (≥ 20 s), or two cough or choking episodes, or three events of any severity.
11.2. Needs Pacing: Pausing of milk flow by the feeder to allow breathing integration.
11.3. Uncoordinated: No pacing is required by the feeder, but the infant is unable to engage in a mature SSB sequence for extended bursts (10 to 30 sucks) with a brief pause between bursts.
11.4. Coordinated: Mature, rhythmic sequence of suck/swallow/breathe at a 1:1:1 or 2:1:1 ratio for extended sucking bursts (10 to 30 sucks).
12. TOTAL ORAL SKILLS (NS): STRESS CUES: Behavioral signs of overstimulation exhibited by an infant, including but not limited to hiccups, yawning, splayed fingers, eyebrow-raising, and saluting.
12.1. Shut Down: Rapid loss of arousal and engagement in response to stress.
12.2. Frequent (stress cues): Abundant stress cues during interaction
12.3. Sometimes (stress cues): Occasional stress cues during an interaction.
12.4. None (stress cues): No stress cues are seen.

Table 2. SMART tool Glossary.

Study design

A multicenter, prospective, cross-sectional, observational study was conducted at four level 3 NICUs in the Chicagoland area. A convenience sample of infants who met the study criteria was prospectively recruited over time at each site during the study period. This study was conducted in accordance with the principles of the Declaration of Helsinki. All methods were performed according to the relevant guidelines and regulations. Institutional Review Board (Advocate Aurora Health IRB #21.001E) protocol approval was obtained, and the study was registered in ClinicalTrials.gov (ID: NCT05729828). The study was conducted according to the protocol. The study protocol was explained to the legal guardians, and informed consent was obtained. The goal of the validity studies was to conduct one feeding observation per infant using a cross-sectional design, with a trained scorer completing both the NEOA and SMART tool assessments based on their observation of the oral feeding. The inter-rater reliability studies involved observation by two observers for the same feeding event. The intra-rater reliability studies had more than one observation per infant by the same observer, six hours apart. Validity tests were conducted at four locations, while reliability tests were conducted at one location. Inclusion criteria included (1) medically cleared for oral feedings and at least at 33 weeks, 0 days post-menstrual age, (2) ability to tolerate enteral feedings, and (3) parental consent available. Exclusion criteria included respiratory support of more than two liters per minute of oxygen via nasal cannula.

There were four sites, each with a site lead. One site had two site leads, making a total of five site leads. The site leads completed the Collaborative Institutional Training Initiative (CITI Program) certification before the start of the study to maintain standards in research, ethics, compliance, and safety. The site leads were involved in study participant identification, recruitment, obtaining consent, data submission to the electronic portal, and ensuring data quality at each site. Additionally, a total of 26 scorers were present from all sites. The scorers included NICU nurses, lactation consultants, and speech-language pathologists. Four hours of standard education and training on the study protocol, NEOA, and SMART tool were provided virtually. All scorers and site leads were required to score three additional feedings using both tools to assess accuracy. Feedback and peer coaching were provided to reach the desired fidelity of feeding skill assessment across scorers. Trained scorers strictly observed feedings and provided no feedback to feeders during the study. The scorer initially documented each feeding observation on paper, and then the site lead transferred the data to an online software form for future analysis using the REDCap software³¹. The scores were automatically calculated online.

Statistical analysis

The SMART tool's psychometric properties were evaluated using "Consensus-based Standards for the Selection of Health Status Measurement Instruments Criteria" (COSMIN Criteria)¹⁴. The plan for psychometric evaluation was to collect evidence for general and measurement properties (Supplementary Table S2). Data capture was done using REDCap³² Software hosted at Advocate Health. Data management and analysis were conducted with SAS statistical software (Version 9.4; SAS Institute, Cary, NC) and GraphPad Prism software (Version 10; Boston, MA). All analyses were performed by research personnel employed by the health system. For all statistical estimates, $p < 0.05$ was considered statistically significant. According to COSMIN standard criteria, to establish content validity, it is estimated that at least seven observations should be tested per item to see a proper difference, not due to chance. Since this tool has ten questions (including five items and five subitems), the plan was to observe at least seventy feeding events for validity testing³³.

The psychometric properties were evaluated using four types of tests: content validity, criterion validity, convergent validity, and reliability³⁴. Psychometric properties are statistical properties of the qualities of a tool. Psychometric properties are not inherent to the tool; rather, they depend on how the tool's scores are interpreted to estimate the construct¹⁴. The interpretation of scores depends on the tool, the patient, the context, the clinician's observations, and their biases. If the clinician administers the tool in a non-standard manner, the validity and reliability of the results will be compromised. For example, jaundice estimation by visual inspection was established in 1969 for white, full-term infants, known as Kramer's rule of jaundice cephalocaudal progression³⁵. However, this score is inaccurate in cases of dark-skinned infants, premature infants, variable progression of jaundice, various etiologies, or severe jaundice³⁶. The original scores, determined by Kramer's rule, were not intended to apply to all populations. Hence, the validity and reliability depend on the interpretation of the scores in each context. Clarity of procedures, glossary, and decision-making algorithms will minimize errors. Construct is the concept that a test is designed to measure. Conceptualizing a construct changes what, why, and how it is measured. Validity refers to the degree to which evidence and theory support the interpretations of test scores for proposed uses of tests. Validation is a joint responsibility of the test developer and test user¹⁴.

- Content validity: Content validity evaluates the relation between test content and the construct it intends to measure. There was no patient data collection in content validity tests. The content validity ratio (CVR) and content validity index (CVI) were calculated for content validity^{37,38}. SMEs independently evaluated each item of the SMART tool. An online survey was conducted using Qualtrics software, and data analysis was performed using Microsoft Excel software. The item has an acceptable level of statistical significance for the CVR and CVI if the calculated number is greater than the minimum required threshold based on the number of experts³⁷.
- Criterion validity: Criterion validity compares the performance of a new tool against a similar available tool with established psychometric properties. For criterion validity¹⁴, the SMART tool was compared to a benchmark tool, the Neonatal Eating Outcome Assessment (NEOA) tool. Due to the detailed design, robust psychometric properties, and the ability to get an FSA score at term equivalency, NEOA was used as a benchmark tool for validity testing. We obtained permission from the authors of NEOA to consider it a comparator tool for research purposes. The post-feed SMART tool total scores were compared to NEOA tool total scores

- at term equivalence. For criterion validity, the results were grouped into a binary option for each tool to generate a 2 × 2 contingency table. The groups were “normal (77 to 90)” and “not normal (18 to 76)” for NEOA, and the groups were “capable (91 to 100)” and “not capable (25 to 90)” for the SMART tool. For dichotomous scores, diagnostic tests included the calculations of sensitivity, specificity, positive predictive value, and negative predictive value³⁹. Spearman’s Rho was used to calculate the correlation strength between the two variables. Continuous scores were summarized using the area under the receiver operator characteristic curves⁴⁰.
- Convergent validity: Construct validity describes the relationship between the tool and other tools measuring the same construct. For convergent validity, we proposed that the scores on both tools (SMART and NEOA) would improve with advancing gestational age. To establish convergent validity, Spearman’s Rho correlation coefficient was used to establish the correlation between the Post-Feed Total scores from the SMART tool and the NEOA tool total score in different groups based on birth gestational age. The term equivalency for NEOA tool scoring was used.
 - Reliability: Reliability or precision refers to the consistency of scores across instances of testing procedures³⁴. Pre-feed and post-feed scores from the SMART tool were compared in reliability studies. Inter-rater reliability was measured by two scorers observing one infant feeding session and scoring using the SMART tool. Intra-rater reliability was calculated with one scorer using the SMART tool to observe two distinct infant feeding sessions on the same infant six hours apart. Reliability studies for the SMART tool provided paired scores. All scores were converted to respective categories based on the decision tree. The categories were then compared to check the agreement between the two scores. McNemar’s test was used to test the symmetry of a paired table for two categories, and Bowker’s test was used for three categories. Cohen’s kappa statistic was used to test the agreement of SMART categories.

Results
Observations

Seventy-six infants were enrolled in the study (Table 3) between May 5, 2021, and November 23, 2021. The median birth gestational age was 33 weeks (range 25 to 39 weeks), with a median birth weight of 2030 g (range 600 to 5250 g). Out of 143 feeding observations, 7 had partial scores (excluded), and 136 had complete scores. The distribution of the 136 complete scores was validity (94 scores), inter-rater (20 scores on 10 feedings), and intra-rater (22 scores on 12 feedings, with two pre-feed scores without a matching post-feed score). Baseline vitals (*n* = 136) described in median (range) were heart rate 161 (128 to 200), Respiratory rate 46 (24 to 64), and oxygen saturation 100% (88 to 100%). Interventions implemented by the feeder include the use of various nipple

Demographics	Infants	%
Infants Enrolled in Tests	76	100
Validity	59	78
Reliability Inter-Rater	8	11
Reliability Intra-rater	9	11
Infant Characteristics	76	100
Birth Gestational Age Groups		
A (22 to 26)	2	3
B (27 to 31)	25	33
C (32 to 36)	39	51
D (37 to 41)	10	13
Cesarean	50	66
Female Sex	35	46
Singleton	52	68
APGAR Under five at 5 min	0	0
Orofacial Issues	1	1
Tongue Tie	2	3
Airway Issues	1	1
Intrauterine Drug Exposure	3	4
Severe Gastrointestinal Issues	1	1
Severe Neurological Issues	3	4
Maternal Characteristics	76	100
Hypertension	38	50
Diabetes	14	18
Obesity	33	43

Table 3. Demographics.

Number of experts = 14	Content validity ratio (CVR) The item is Essential		Content validity index (CVI) The Item is Related	
	Required	Actual	Required	Actual
Item				
State	0.51	1	0.8	1
Motor Tone	0.51	0	0.8	1
Autonomic Instability	0.51	1	0.8	1
Response to Stimulation	0.51	1	0.8	1
Total Oral Skills	0.51	0.57	0.8	1

Table 4. Evidence of content validity based on consensus of subject matter Experts.

Number of feeding observations (N)		NEOA tool			Diagnostic tests	Correlation	95% CI
		Normal	Not normal	Total			
					Sensitivity	0.76	0.63–0.89
SMART tool	Capable	32	9	41	Specificity	0.82	0.72–0.93
	Not Capable	10	43	53	Positive Predictive Value	0.78	0.65–0.91
	Total	42	52	94	Negative Predictive Value	0.81	0.71–0.92

Table 5. Evidence of criterion validity based on feeding observations (N).

flow rates and the application of other infant-directed feeding techniques, such as swaddling and co-regulated pacing.

COSMIN criteria: general properties

The purpose of the SMART tool is to measure oral feeding skills in infants in the NICU. This study assessed its psychometric properties compared to the NEOA tool. The inclusion criteria for the study were infants above 33 weeks and with low oxygen need (≤ 2 L per minute). The structure of the SMART tool is based on the Synactive Theory of Development²⁵. The SMART tool's design includes a decision tree to signal readiness, safety, and feeding quality, making scores interpretable and clinically relevant. After training, the scorers reported a completion time of less than 5 min for the SMART tool following the oral feeding observation, which makes it highly feasible to integrate into routine clinical workflow. The SMART tool is a multidimensional, formative model (Supplementary Figure S3)⁴¹. All items had moderate to strong correlations with total scores.

COSMIN criteria: measurement properties

- Content Validity (Table 4): 14 SMEs responded to an online survey. All SMEs worked at a level 3 NICU. For CVR, all items were considered *essential* except for the motor tone. The item, motor tone, was not removed from the tool, as it is a subsystem in the Synactive Theory of Development²⁵. In the dimensionality test (Supplementary Figure S3), Motor Tone moderately correlated with Total Score, State, and Response to Stimulation. Total Oral Skills moderately correlated with Autonomic Instability and Response to Stimulation. For CVI, all items were *related* to the tool's purpose.
- Criterion Validity (Table 5): The SMART tool's sensitivity (True positive rate) to detect NEOA normal category cases is 32/42, or 0.76 (95% CI 0.63–0.89). The SMART tool's specificity to detect NEOA non-normal category cases is 43/52, or 0.82 (95% CI 0.72–0.93). There was no “missing feeding observation” data in the validity study. Comparing the SMART tool score ratings and the NEOA tool score ratings, Spearman's Rho of $r = 0.706$ ($n = 94$) indicates a moderate to strong positive correlation of score ratings between the tools, which is statistically significant at $p < 0.0001$, signifying high confidence. The receptor operator characteristic curve (Supplementary Figure S4) is a graphical plot illustrating the diagnostic ability of the SMART tool to discriminate between capable and non-capable cases, as defined by the NEOA tool's normal category⁴². The area under the curve (AUC) of the SMART tool to identify capable cases, defined by a NEOA normal classification, is 0.79 (95% CI 0.73–0.86), bordering excellent⁴⁰.
- Convergent Validity (Table 6): Feeding observations for all gestations improved with increased gestational age, and there was a statistically significant high correlation between the SMART tool and NEOA scores.
- Reliability (Table 7): Inter-rater reliability tests showed high agreement between the two scorers using the SMART tool for observed feeds. Pre-feed inter-rater scores had 80% agreement, and post-feed inter-rater scores had 100% agreement. There were no significant differences in the symmetry of scores. Intra-rater reliability tests had variable results. The pre-feed intra-rater scores were 50% agreeable, whereas the post-feed intra-rater scores were 80% agreeable, likely related to variation in infant performance from feeding to feeding. None of the differences in categories were significant.

Groups by birth gestational age (weeks)	N	Spearman's correlation	95% Confidence interval	p-value
Group A (22 to 26)	2	Not Reported	Not Reported	Not Reported
Group B (27 to 31)	29	0.48	0.13 to 0.73	0.01
Group C (32 to 36)	53	0.79	0.66 to 0.88	< 0.001
Group D (37 to 41)	10	0.65	Not Reported	0.05
All (22 to 39)	94	0.71	0.59 to 0.80	< 0.001

Table 6. Evidence of convergent validity based on feeding observations (N).

Test	Compare scores	N	Agree (%)	Categories	Symmetry	p-value	Weighted kappa (SE)	Kappa (95% CI)
Inter-Rater	Pre-Feed vs. Pre-Feed	10	80	3	Bowker's	0.57	0.72 (0.17)	0.38 to 1
	Post-Feed vs. Post-Feed	10	100	2	McNemar's	-	1 (0)	1 to 1
Intra-rater	Pre-Feed vs. Pre-Feed	12	50	3	Bowker's	0.80	0 (0.22)	-0.43 to 0.43
	Post-Feed vs. Post-Feed	10	80	2	McNemar's	0.15	0.6 (0.23)	0.14 to 1

Table 7. Evidence of reliability based on feeding observations (N). Cohen's kappa statistics are used to test the agreement of different categories. The strength of agreement is described based on the kappa coefficient as follows: poor (0), slight (0.01 to 0.20), fair (0.21 to 0.40), moderate (0.41 to 0.60), strong (0.61 to 0.80), and almost perfect (0.81 to 1).

Discussion
Safety and quality of infant feedings using the SMART tool

Clinicians need a gold-standard FSA tool with well-established psychometric properties to measure and trend the development of infant feeding skills in the NICU. None of the available tools in our literature review met all of the selection criteria^{8,10–13,16}. This study established the validity, reliability, and feasibility of the SMART tool for FSA. These results are highly relevant to bedside clinicians in NICUs, as they allow implementation of the SMART tool to objectively measure infant feeding skill development, including readiness, safety, and quality of every oral feeding. Traditional measurements of feeding competence have relied on volume of intake rather than feeding skill. The SMART tool provides a numeric value to represent feeding skill (quality) in addition to feeding volume (quantity). Human-centered design principles were rigorously considered to increase the feasibility of the SMART tool, such as a focus on the end-user clinicians, interprofessional collaboration, iterative design with feedback, and using a system perspective for easy integration into the workflow⁴³.

A detailed comparison of the SMART tool with other available infant feeding skill assessment tools is provided in Table 8. It demonstrates the robustness of the SMART tool in providing a comprehensive assessment of infant oral feeding skills, being intuitive to understand, feasible, and achieving high diagnostic testing accuracy.

The latent variable evaluated in the SMART tool is feeding skills. Five constructs assess feeding skills, including state, motor tone, autonomic instability, response to stimulation, and total oral skills. These constructs aim to objectively describe the safety and quality of oral feedings in infants and classify the infant's feeding skill level. This aligns with the formative model of measurement research⁴¹. This implies that the individual constructs of the SMART tool cause the latent variable of feeding skills and not the other way around; feeding skills cause the indicators of the five constructs. We observed the multidimensional aspect of the SMART tool. The constructs (state, motor tone, and autonomic instability) correlate moderately with the total score. In contrast, the following constructs (response to stimulation and total oral skills) strongly correlate with the total score. See Supplementary Figure S3 for the Dimensionality Test.

There are several advantages to using an objective, systematic, standard assessment for infant feeding skills. This includes (1) definite criteria that help to identify safety “hard stops” before and during feeding, (2) provide a numeric value to represent feeding skills (quality), in addition to feeding volume (quantity), (3) using the tool to identify the etiology of oral feeding difficulty, (4) creating an individualized feeding plan, (5) monitoring progress over time to determine the effectiveness of feeding interventions, (6) communicating skill development with all involved in the infant's care and family, and (7) potentially using it as a predictive tool regarding long-term developmental outcomes⁸. The proactive approach, focused on the safety and quality of feedings, using a standardized feeding assessment tool, can enhance pleasurable feeding experiences and minimize adverse events that lead to stress and trauma. While we do not recommend for or against using these scores as a basis for NICU discharge, users are encouraged to document the safety and quality of feedings in daily progress notes and discharge summaries to provide continuity of care and assess feeding skill development post-discharge.

Limitations

The SMART tool is intended for breast- and bottle-fed infants. However, only 4% of the study's observations were of directly breastfed infants. This limits the generalization of the results to this patient population, and additional research is needed. Other patient populations not widely represented in the sample size include infants with significant co-morbidities and extremely preterm infants. The tool was designed for infants beginning at 32 weeks post-menstrual age; however, during study protocol development, inclusion criteria were agreed at 33

Name of FSA tool	Infant driven feeding scale (IDFS)	Early feeding skills (EFS)	Neonatal eating outcome assessment (NEOA)	SMART tool (ST)
Design				
First reported (Year)	2007	2005	2018	2025
For Preterm Infants	Yes	Yes	Yes	Yes
For Breast & Bottle fed	Yes	Yes	Yes	Yes
Measure Construct	Readiness and Quality	Feeding Skill	Feeding Skill	Feeding Skill
Underlying Theory	Not Reported	Not Reported	Preemie Neuro	Synactive Development
Framework	Not Reported	Multidimensional	Multidimensional	Multidimensional
Complexity				
Factors	2	5	3	5
Items	2	19	18	5
Sub Items	0	0	0	5
Adjust for volume	No	No	Yes	No
Adjust for prematurity	No	No	Yes	No
Adjust for no vitals	Not Reported	Total Score reduced	Not Reported	Total score is imprecise
Total Score	1 to 5	19 to 57	18 to 90	25 to 100
Clinical Decision Tree	No	No	No	Yes
Detect deficiency area	No	Yes	Yes	Yes
Non-Scoring Items	1	0	26	12
Complexity summary	Low	Medium	High	Medium
Skill Level (Total Score)				
Poor	5	Not Reported	Feeding Challenge (18 to 57)	Caution (25 to 59)
Fair	2 to 3	Not Reported	Questionable (58 to 76)	Developing (60 to 90)
Excellent	1	Not Reported	Normal (77 to 90)	Capable (91 to 100)
Validity Tests				This Study
Content	Consensus (Delphi)	Yes	Yes	Independent Agreement CVR = 1 (goal > 0.5) CVI = 0.57 (goal > 0.8)
Criterion	Not Reported	IDFS (Quality)	NOMAS (term)	NEOA (goal > 0.70) Sensitivity = 0.76 Specificity = 0.82
Convergent	Not Reported	Improve with Gestational Age at birth. Pearson correlation = 0.23	Not Reported	Improve with Gestational Age at birth. Spearman's correlation = 0.71
Structural	Not Reported	Strong factor analysis	Not Reported	Not Reported
Cross-culture	Translations done	Translations done	Not Reported	Not Reported
Reliability Tests				This Study
Inter-rater (post feed)	Not Reported	Acceptable	High (correlation = 0.9) Fleiss's kappa = 0.56)	Agreement = 100% Cohen's kappa = 1)
Intra-rater (post-feed)	Not Reported	Not Reported	Not Reported	Agreement = 80% Cohen's kappa = 0.6
Internal Consistency	Not Reported	High	Not Reported	Not Reported
Feasibility				
Cost (License to use)	Yes	Yes	No	No
Training	Yes	Yes	Yes	Yes
Electronic integration	Yes	No	No	Notes, Tables, Trends
Time needed (minutes)	5	10	20	5

Table 8. Summary of psychometric properties of infant feeding skill assessment tools. NOMAS is the neonatal Oral-Motor assessment scale (NOMAS) for infant oral-motor skills assessment.

weeks due to inter-site variations in feeding practice. Due to COVID-19, there were limitations, such as low census, slow enrollment, learning attrition among scorers, and variable learning with online training. This may have led to inter-site variations in score interpretations, called *construct-irrelevant variance*. Reliability tests were conducted at only one site to overcome the limitations during COVID-19. Few measurement properties under COSMIN criteria were not measured as they do not apply to the SMART tool because the SMART tool is multidimensional; hence, inter-item correlations are low, and structural validity and internal consistency reliability tests are not applicable.

Conclusion

The results of this study are highly relevant to all NICU clinicians. The study established the validity, reliability, and feasibility of the SMART tool for FSA, allowing this novel tool to become the “standard FSA tool” NICU clinicians use to objectively evaluate feeding skills beyond the volume of oral intake across this health system. It meets all of the criteria that our health system was aiming for in the search for a standard FSA tool: (1) application: use for infants (term and premature) with different manners of intake (breast and bottle), (2) holistic: comprehensive in assessing the many facets of feeding, (3) interpretability: provides a score for interpretation regarding safety and quality of feeds that can be communicated between clinicians and families, (4) robust psychometric properties and (5) feasible: readily accessible and efficient for bedside clinicians to complete training on the tool, perform scoring, interpret results document in the electronic health record and at a low cost to the health system. The results of this study were presented at national conferences^{44,45}. We have implemented this tool in our health system based on its strong psychometric properties. Future studies will evaluate the SMART tool’s effect on clinical outcomes.

Data availability

The datasets generated during and/or analyzed during the current study are available from the corresponding author upon reasonable request.

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Author contributions

R.M., the primary investigator, conceptualized and designed the SMART Tool, designed the study and data collection instruments, coordinated and supervised data collection, carried out the initial analyses, created all figures and tables, drafted the initial manuscript, and critically reviewed and revised it. A.A., the sub-investigator, designed the SMART Tool, designed the study, coordinated and supervised data collection, provided direct education, drafted the initial manuscript, and critically reviewed and revised the manuscript. C.L., the sub-investigator, designed the study, coordinated system approvals, coordinated and supervised data collection, provided direct education, drafted the initial manuscript, and critically reviewed and revised the manuscript. All authors approved the final manuscript as submitted and agreed to be accountable for all aspects of the work.

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

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